Vision-Language Models in Radiology: Zero-Shot Classification, Cross-Modal Retrieval, and Factual Radiology Report Generation

Research Proposal for Biomedical Image Analysis and Processing

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Abstract

Radiology report generation and interpretation are critical yet time-consuming tasks in clinical workflows. Recent advances in vision-language models (VLMs), particularly contrastive learning approaches like CLIP, have demonstrated promising capabilities for zero-shot medical image classification and cross-modal retrieval. However, significant challenges remain: (1) factual hallucination in generated reports, where models produce clinically incorrect findings; (2) limited cross-institutional robustness, with performance degradation on external datasets; and (3) lack of uncertainty quantification for clinical deployment. This research proposes a comprehensive study of VLMs for radiology, focusing on three interconnected tasks: zero-shot pathology classification, image-text retrieval, and factuality-aware report generation. Building on CheXzero and R2Gen as foundational architectures, we propose novel extensions incorporating uncertainty quantification, RadGraph-based factuality constraints, and systematic cross-dataset evaluation. Using publicly available datasets (IU X-Ray, NIH ChestX-ray14), we aim to develop methods that advance both technical performance and clinical deployability. This work is designed as a graduate course project with clear pathways to publication at MICCAI, MIDL, or related venues, addressing critical gaps in medical AI research.

Contents

1 Introduction

1.1 Clinical Motivation

Radiology is the cornerstone of modern medical diagnosis, with over 3 billion imaging studies performed annually worldwide. Each chest X-ray examination requires expert interpretation to detect pathologies ranging from common conditions (pneumonia, pleural effusion) to life-threatening emergencies (pneumothorax, aortic dissection). However, the radiology workforce faces unprecedented challenges:

- **Global radiologist shortage:** The World Health Organization estimates a deficit of over 2 million radiologists globally, particularly severe in low- and middle-income countries.
- **Increasing imaging volumes:** Annual growth rate of 5-10% in imaging studies outpaces radiologist training capacity.
- **Diagnostic errors:** Studies report 3-5% error rates in radiology interpretation, with significant consequences for patient outcomes.
- **Workflow inefficiencies:** Radiologists spend 60-70% of their time on report writing rather than complex diagnostic reasoning.

Artificial intelligence, specifically deep learning for medical image analysis, has emerged as a promising solution. Early successes in classification tasks (e.g., CheXNet achieving radiologist-level pneumonia detection) demonstrated technical feasibility. However, clinical translation remains limited due to fundamental challenges in trustworthiness, generalizability, and practical deployment.

1.2 The Promise of Vision-Language Models

Recent advances in vision-language models (VLMs) offer a paradigm shift from traditional supervised learning. Instead of requiring thousands of manually labeled examples per disease, VLMs learn joint representations of images and text through self-supervised contrastive learning on naturally occurring image-report pairs. This approach provides several advantages:

1. **Zero-shot learning:** Models can classify new diseases without disease-specific training data, using only text descriptions.
2. **Data efficiency:** Self-supervised pretraining on unlabeled image-text pairs reduces annotation burden.
3. **Unified framework:** Single model supports multiple tasks (classification, retrieval, generation).
4. **Interpretability:** Text grounding provides clinically meaningful explanations.

Models like CLIP (Contrastive Language-Image Pre-training) have revolutionized natural image understanding. Medical adaptations—CheXzero, GLORIA, BioViL—demonstrate that similar principles apply to radiology, achieving expert-level performance on pathology detection tasks.

1.3 Critical Research Gaps

Despite promising results, current medical VLMs face severe limitations that impede clinical deployment:

Gap 1: Factual Hallucination
VLMs trained on report generation frequently produce fluent but clinically incorrect text. A model might generate "fracture of the left 4th rib" when no fracture exists, or omit critical findings like pneumothorax. Traditional NLG metrics (BLEU, ROUGE) fail to detect these errors, as they measure linguistic similarity, not clinical accuracy.

Gap 2: Cross-Institutional Brittleness
Models trained on single-institution data (e.g., MIMIC-CXR from Beth Israel Deaconess) show performance degradation of 5-15% AUROC when tested on external hospitals. Contributing factors include equipment differences, population demographics, and reporting style variations.

Gap 3: Lack of Uncertainty Quantification
Clinical decisions require knowing when a model is uncertain. Current VLMs produce point predictions without confidence estimates, making it impossible to identify out-of-distribution cases or defer to human experts appropriately.

Gap 4: Limited Grounding
Many models fail to link generated findings to image regions. Without visual evidence for each claim, radiologists cannot verify model outputs or understand failure modes.

Gap 5: Evaluation Misalignment
Standard metrics (BLEU, ROUGE) correlate poorly (r=0.3) with clinical utility. Emerging alternatives like RadGraph F1 and CheXbert accuracy better capture clinical correctness but remain underutilized in training objectives.

1.4 Research Objectives

This project aims to address these gaps through three primary objectives:

Objective 1: Develop uncertainty-aware zero-shot classification
Extend CheXzero with Monte Carlo Dropout and conformal prediction to provide calibrated confidence intervals, enabling appropriate human-AI collaboration.

Objective 2: Create factuality-constrained report generation
Integrate RadGraph-based supervision into R2Gen training, penalizing entity hallucination and rewarding grounded generation.

Objective 3: Establish cross-dataset robustness benchmarks
Systematically evaluate model performance across multiple public datasets (IU X-Ray, NIH ChestX-ray14), analyze failure modes, and propose domain adaptation techniques.

1.5 Scope and Constraints

This project focuses on **publicly available resources only**:

- **Datasets:** IU X-Ray (3,955 studies), NIH ChestX-ray14 (112,120 images)
- **Modality:** Chest X-ray (frontal view)
- **Tasks:** Classification, retrieval, report generation
- **Compute:** Single GPU experiments (NVIDIA V100 or A100)

While MIMIC-CXR (227K studies) offers larger scale, its credentialing requirements and 1-2 week access delay make it impractical for an 8-month course project. Our approach demonstrates that meaningful research contributions are possible with smaller, fully public datasets by focusing on methodological innovation rather than scale.

1.6 Expected Contributions

This research will deliver:

1. **Technical contributions:**
    - Uncertainty-aware VLM architecture with calibrated predictions
    - RadGraph-guided training loss for factual report generation
    - Cross-dataset evaluation protocol and benchmark results
2. **Scientific contributions:**
    - Systematic analysis of VLM robustness across datasets
    - Quantification of factuality improvement via knowledge graph supervision
    - Ablation studies isolating key design choices
3. **Practical contributions:**
    - Open-source implementation with pretrained weights
    - Reproducible evaluation framework
    - Clinical error analysis and failure mode taxonomy
4. **Publication:**
    - Target venue: MICCAI, MIDL, ML4H Workshop, or TMI journal
    - Estimated timeline: Submission by Month 7-8

1.7 Document Organization

The remainder of this proposal is organized as follows: Section 2 reviews related work in medical VLMs, report generation, and evaluation methods. Section 3 describes datasets and their characteristics. Section 4 presents our proposed methodology with three research extensions. Section 5 details the experimental plan and evaluation protocol. Section 6 discusses expected contributions and limitations. Section 7 addresses ethical considerations. Section 8 provides a detailed timeline. Section 9 concludes with future directions.

2 Related Work

2.1 Contrastive Vision-Language Learning

2.1.1 Natural Image VLMs

The CLIP model [?] pioneered large-scale contrastive learning between images and text, demonstrating remarkable zero-shot transfer to downstream tasks. Trained on 400M image-text pairs from the internet, CLIP learns to align visual and language representations in a shared embedding space. The key insight: maximizing agreement between positive pairs (matching image-text) while minimizing agreement with negative pairs (non-matching) creates semantically meaningful representations.

CLIP's architecture consists of:

- **Image Encoder:** Vision Transformer (ViT) or ResNet
- **Text Encoder:** Transformer (GPT-style)
- **Training Objective:** Symmetric cross-entropy over cosine similarities

Zero-shot classification emerges naturally: given image I and candidate labels $\{L_1, ..., L_N\}$, embed each as prompt "a photo of [label]", compute $\text{sim}(I, L_i)$, and select $\arg\max_i \text{sim}(I, L_i)$.

However, direct application of CLIP to medical imaging fails due to domain mismatch: (1) medical images differ fundamentally from natural photos, (2) clinical terminology absent from internet text, and (3) diagnostic reasoning requires fine-grained discrimination beyond object recognition.

2.1.2 Medical VLM Adaptations

ConVIRT [?] was among the first to adapt contrastive learning to radiology. Pretrained on MIMIC-CXR's 377K image-report pairs, ConVIRT achieves +3.2% AUROC improvement over ImageNet initialization on CheXpert classification. Key findings: (1) medical-specific pretraining outperforms natural image pretraining, (2) gains largest in few-shot regimes (1-10% labeled data), (3) transfer learning effective across chest X-ray datasets.

CheXzero [?] demonstrates expert-level zero-shot pathology detection. Using ResNet-50 + BioClinicalBERT, CheXzero achieves 0.873 mean AUROC on 5 CheXpert diseases without disease-specific labels. Critical innovation: carefully engineered clinical prompts ("Findings consistent with pneumonia" vs. "No evidence of pneumonia"). Limitations: prompt engineering requires clinical expertise and manual trial-and-error.

GLORIA [?] extends global contrastive learning with local attention, matching image regions to report sentences. Achieves state-of-the-art retrieval (32.2% Recall@1 on MIMIC-CXR) and provides interpretable attention maps. Local alignment enables grounding but increases computational cost (4× V100 GPUs, 2-3 days training).

BioViL [?] (Microsoft, ECCV 2022) uses RadGraph to create structured image-text alignments during pretraining. By parsing reports into knowledge graphs and aligning entities to image regions, BioViL improves phrase grounding accuracy by 20% over baseline. Demonstrates value of structured medical knowledge in VLM training.

MedCLIP, PubMedCLIP, PMC-CLIP explore pretraining on broader medical data (PubMed figures, captions) for multi-domain transfer. While achieving reasonable zero-shot performance across modalities (CT, MRI, microscopy), they lack radiology-specific optimization of specialized models like CheXzero.

2.2 Radiology Report Generation

Automatic report generation aims to produce coherent, clinically accurate radiology reports from images. Unlike image captioning, medical reports require:

- Structured format (Findings, Impressions, Comparisons)
- Clinical precision (diagnostic terminology, negation handling)
- Evidence linking (findings supported by visible pathology)
- Absence documentation ("no evidence of X" equally important)

2.2.1 Encoder-Decoder Architectures

Early approaches adapted Show-Attend-Tell [?] from image captioning. **TieNet** [?] uses multi-task learning (generation + classification) to improve clinical accuracy. **AdaAtt** applies adaptive attention to focus on relevant regions. These methods achieve BLEU-4 $\approx$ 0.18-0.20 on IU X-Ray but suffer from generic descriptions and factual errors.

R2Gen [?] introduces a relational memory module storing clinical patterns. Memory slots specialize to common findings ("lungs clear", "cardiomegaly"), improving coherence. Achieves

BLEU-4 = 0.203 on IU X-Ray, CIDEr = 0.485. Multi-scale visual encoding (CNN + Transformer) captures both global and local features. Despite strong linguistic metrics, R2Gen lacks explicit factuality constraints.

CMN (Clinical Memory Network) [?] extends memory-based generation with clinical prototypes retrieved from training data. Achieves modest CheXbert F1 improvement but limited evaluation of factual correctness.

PPKED [?] integrates medical knowledge graphs (RadLex, UMLS) to guide generation. While conceptually appealing, complex pipeline and limited code availability hinder reproducibility.

2.2.2 Retrieval-Based and Hybrid Methods

RATCHET [?] retrieves similar historical reports and adapts them to current image. Achieves high fluency but limited novelty—struggles with rare findings absent from retrieval database.

Recent trends (2023-2024): Large language model (LLM) integration. LLaVA-Med, Med-Flamingo, and RadFM use instruction-tuned LLMs (LLaMA, GPT-3.5) conditioned on visual features. While impressive in qualitative examples, quantitative factuality evaluation remains limited.

2.3 Factuality and Grounding

2.3.1 Evaluation Beyond BLEU/ROUGE

Traditional NLG metrics (BLEU, ROUGE, METEOR, CIDEr) measure n-gram overlap with reference text. Critical flaw: high scores possible with clinically incorrect content. Example: "mild cardiomegaly" vs. "severe cardiomegaly" have high BLEU (8/9 words match) but opposite clinical meaning.

RadGraph [?] extracts clinical entities and relations from reports, converting text to knowledge graphs. RadGraph F1 metric compares hypothesis and reference graphs, capturing clinical content independently of phrasing. Correlation with expert assessment: r=0.7 (vs. BLEU r=0.3). Now standard evaluation for medical report generation.

CheXbert [?] fine-tunes BERT to classify 14 CheXpert labels from report text. CheXbert accuracy measures whether generated reports convey correct diagnostic findings. Simpler than RadGraph but limited to 14 binary labels.

2.3.2 Grounding and Attention

MS-CXR [?] provides sentence-to-region annotations, enabling supervised grounding. Models learn to link "opacity in left lower lobe" to specific image coordinates.

GREEN [?] uses entity-level supervision to reduce hallucination. By explicitly annotating medical entities (anatomy, observations) and penalizing ungrounded generation, GREEN achieves 15% reduction in false entity rate.

FactualGAN [?] trains adversarial discriminator to distinguish factual from hallucinated reports, providing training signal for generator.

2.4 Cross-Dataset Evaluation and Domain Shift

Most medical VLM papers evaluate on a single dataset (typically MIMIC-CXR or IU X-Ray). Limited cross-dataset validation obscures generalization failure. Known issues:

- **Equipment variation:** GE vs. Siemens scanners produce different image characteristics
- **Population demographics:** Age, comorbidity distributions vary across institutions

- **Reporting style:** Phrase choices, sentence structure, abbreviations differ
- **Disease prevalence:** COVID-19, endemic diseases, referral patterns shift distributions

Few studies systematically evaluate robustness:

- CheXzero tests on PadChest (Spain) and observes 3-5% AUROC drop
- GLORIA shows modest degradation on NIH-14
- Most papers ignore external validation entirely

Research gap: Comprehensive cross-dataset benchmarking with analysis of failure modes and domain adaptation techniques.

2.5 Uncertainty Quantification in Medical AI

Clinical deployment requires knowing *when* a model is uncertain. Uncertainty quantification (UQ) methods:

- **Bayesian Deep Learning:** Monte Carlo Dropout [?], variational inference
- **Ensemble Methods:** Train multiple models, report disagreement
- **Conformal Prediction:** Distribution-free confidence sets with coverage guarantees
- **Calibration:** Temperature scaling, Platt scaling for probability calibration

Medical imaging applications:

- Dermoscopy: Uncertainty-aware skin lesion classification defers 15% of ambiguous cases to dermatologists, improving overall accuracy
- Pathology: Conformal prediction provides slide-level confidence intervals for tumor classification
- Radiology: Limited work on VLM uncertainty—most models produce overconfident predictions

Research gap: Integration of principled UQ into medical VLMs for safe clinical deployment.

2.6 Summary and Positioning

Existing work demonstrates:

- Contrastive VLMs achieve strong zero-shot classification (CheXzero: 0.87 AUROC)
- Report generation reaches moderate linguistic quality (R2Gen: BLEU-4 = 0.20)
- Clinical evaluation metrics (RadGraph, CheXbert) quantify factual correctness

Critical gaps remain:

- Factual hallucination unaddressed in training objectives
- Cross-dataset robustness poorly understood
- Uncertainty quantification absent

- Grounding between generated text and image regions weak

Our contribution: This project bridges these gaps by:

1. Incorporating RadGraph-based factuality loss in report generation training
2. Systematic cross-dataset evaluation (IU X-Ray ↔ NIH-14)
3. Uncertainty quantification via MC Dropout and conformal prediction
4. Public dataset focus, enabling reproducible research

Next section describes datasets in detail.

3 Datasets

This section provides comprehensive descriptions of publicly available datasets used in our research. All datasets require no institutional access or credentialing beyond simple registration (if any).

3.1 IU X-Ray (Indiana University Chest X-Ray Collection)

3.1.1 Overview

- **Institution:** Indiana University School of Medicine
- **Year:** 2015
- **Access:** Fully public via OpenI https://openi.nlm.nih.gov/
- **Size:** 7,470 images, 3,955 studies, 3,955 patients
- **File Size:** ~7 GB images + 5 MB reports (XML)

3.1.2 Data Composition

Images:

- Modality: Chest X-ray (frontal PA/AP + lateral)
- 40% frontal-only, 60% frontal+lateral
- Average: 1.89 images per study
- Format: PNG, grayscale, resolution 2048×2048 to 3000×2500

Reports: Each study has structured radiology report with 4 sections:

1. **Comparison:** Reference to prior studies (if available)
2. **Indication:** Clinical reason for examination
3. **Findings:** Detailed observations of chest anatomy and pathology
4. **Impression:** Summary diagnosis and recommendations

Average report length: 35-40 words. Characteristics: relatively concise, high frequency of negation ("no acute disease"), some PHI redacted with placeholders.

Annotations:

- Manual expert labels for 14 observations: Normal (28.1%), Cardiomegaly (9.5%), Lung Opacity (16.6%), Edema (2.2%), Consolidation (4.4%), Pneumonia (4.0%), Atelectasis (5.1%), Pneumothorax (1.1%), Pleural Effusion (7.7%), etc.
- Severity grades: Mild, Moderate, Severe
- MeSH (Medical Subject Headings) tags: Controlled vocabulary for retrieval

3.1.3 Data Splits

Standard split used in literature (e.g., R2Gen, CMN):

Split	Studies	Percentage
Train	2,770	70%
Validation	585	15%
Test	600	15%

Random split at study level. We adopt this standard for comparability.

3.1.4 Strengths and Limitations

Strengths:

- Only large-scale public dataset with paired images and reports
- Well-structured reports enable easy parsing
- Manual disease annotations provide gold-standard labels
- Standard benchmark for report generation (>100 papers)
- Multi-view data (frontal + lateral) provides richer information

Limitations:

- Small scale (3,955 vs. MIMIC-CXR's 227K) limits model capacity
- Single institution → potential bias in equipment, population, reporting style
- No demographic data (age, gender, race) → cannot study fairness
- Short reports (35 words) vs. typical clinical reports (50-100 words)
- Class imbalance: rare diseases like pneumothorax (1.1%) underrepresented

3.1.5 Use in This Project

Primary dataset for:

- Training vision-language models (contrastive pretraining)
- Report generation experiments (R2Gen baseline and extensions)
- Image-text retrieval evaluation

Preprocessing:

1. Parse XML to extract findings and impression sections
2. Resize images to 224×224 (CheXzero) or 320×320
3. Normalize with ImageNet statistics or dataset-specific mean/std
4. Tokenize text with BioClinicalBERT tokenizer (max 128 tokens)

3.2 NIH ChestX-ray14

3.2.1 Overview

- **Institution:** National Institutes of Health Clinical Center
- **Year:** 2017
- **Access:** Fully public https://nihcc.app.box.com/v/ChestXray-NIHCC
- **Size:** 112,120 images from 30,805 patients
- **File Size:** ~45 GB (uncompressed PNG)

3.2.2 Data Composition

Images:

- Modality: Chest X-ray (frontal view only: PA/AP)
- Format: PNG, 1024×1024, 8-bit grayscale
- Standardized resolution (unlike IU X-Ray's variable sizes)

Labels: 14 disease categories extracted via NLP from original radiology reports (reports not publicly released):

Disease	Positive Cases	Prevalence
Infiltration	19,894	17.7%
Effusion	13,317	11.9%
Atelectasis	11,559	10.3%
Nodule	6,331	5.6%
Pneumothorax	5,302	4.7%
Mass	5,782	5.2%
Consolidation	4,667	4.2%
Pleural Thickening	3,385	3.0%
Cardiomegaly	2,776	2.5%
Emphysema	2,516	2.2%
Edema	2,303	2.1%
Fibrosis	1,686	1.5%
Pneumonia	1,431	1.3%
Hernia	227	0.2%
No Finding	60,361	53.8%

Important: Labels are *automatically extracted via NLP*, not manually annotated. Estimated 10-20% label noise due to negation errors, context misunderstanding, and mention vs. diagnosis ambiguity.

Bounding Boxes: Subset of 880 images have radiologist-drawn bounding boxes for 8 diseases (Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax). Enables weakly supervised localization.

3.2.3 Data Splits

Common split:

Split	Images	Percentage
Train	70,000	62%
Validation	17,120	15%
Test	25,000	23%

Note: Random split at image level (not patient level). Some papers use patient-level splits to avoid data leakage.

3.2.4 Strengths and Limitations

Strengths:

- Large scale (112K images) enables robust deep learning
- Fully public with no access barriers
- Standardized 1024×1024 resolution simplifies preprocessing
- Widely used benchmark (>500 citing papers)
- Bounding box subset enables localization research
- Diverse patient population (NIH Clinical Center)

Limitations:

- **No reports:** Cannot use for report generation or VLM pretraining
- Label noise (10-20% error rate) due to NLP extraction
- Extreme class imbalance (Hernia: 0.2% vs. No Finding: 53.8%)
- Frontal view only (no lateral)
- Single institution bias
- No demographic data in public release

3.2.5 Use in This Project

Primary use cases:

- **Zero-shot classification evaluation:** Test VLMs pretrained on IU X-Ray without fine-tuning on NIH-14. Measures cross-dataset generalization.
- **Cross-dataset robustness:** Train on IU X-Ray (with reports), test on NIH-14 (classification only). Quantify performance drop.
- **Weakly supervised localization:** Evaluate attention maps on bounding box subset.

Not suitable for:

- Report generation (no reports available)
- Image-text retrieval (no text beyond 14 labels)

Table 1: Comparison of Public Chest X-Ray Datasets

Dataset	Images	Reports	Labels	Access	Best For
IU X-Ray	7.5K	✓	Manual	Public	Generation
NIH-14	112K	×	NLP	Public	Classification
MIMIC-CXR	377K	✓	NLP	Credentialed	Pretraining
CheXpert	224K	×	NLP	Registration	Multi-label
PadChest	160K	Spanish	174 labels	Public	European

3.3 Dataset Comparison and Selection Rationale

Table 1 compares key characteristics:

Why IU X-Ray + NIH-14?

1. **Complementary:** IU X-Ray provides reports for generation; NIH-14 provides scale for classification
2. **Fully public:** No credentialing delays or usage restrictions
3. **Standard benchmarks:** Both widely used, enabling comparison with prior work
4. **Sufficient for graduate project:** Combined 115K images adequate for meaningful experiments

Limitations accepted:

- Cannot match performance of models trained on MIMIC-CXR (10× larger)
- Must frame contributions as methodological innovations rather than state-of-the-art performance
- Focus on robustness, factuality, and uncertainty rather than raw accuracy

3.4 Ethical Considerations for Datasets

Both datasets de-identified per HIPAA standards:

- Patient names, dates, IDs removed
- Ages >89 redacted to "90+"
- Geocodes broader than state level removed

Institutional Review Board (IRB) approval obtained by original dataset creators. Our use constitutes secondary analysis of public data, typically exempt from additional IRB review.

Potential biases:

- Single-institution datasets may not generalize to global populations
- Referral patterns at academic medical centers differ from community hospitals
- Unknown demographic distributions limit fairness analysis

We acknowledge these limitations and will discuss generalization constraints in our final paper.

4 Proposed Methodology

4.1 Overview

Our approach combines two foundational architectures:

1. **CheXzero** [?]: Contrastive vision-language model for zero-shot classification and retrieval
2. **R2Gen** [?]: Memory-driven transformer for report generation

We propose three novel extensions addressing critical gaps:

- **Extension 1 (Classification):** Uncertainty-aware zero-shot classification via Monte Carlo Dropout and conformal prediction
- **Extension 2 (Generation):** Factuality-constrained report generation using RadGraph-based training loss
- **Extension 3 (Robustness):** Cross-dataset evaluation protocol with systematic failure analysis

Each extension is designed to be:

- **Feasible:** Implementable within 2-3 months on single GPU
- **Novel:** Addresses underexplored research gap
- **Publishable:** Sufficient contribution for MICCAI/MIDL workshop or conference paper

4.2 Baseline: CheXzero for Zero-Shot Classification

4.2.1 Architecture

CheXzero consists of:

Image Encoder f_v:

- ResNet-50 pretrained on ImageNet
- Input: $x \in \mathbb{R}^{3\times320\times320}$ (chest X-ray, replicated to 3 channels)
- Output: $v = f_v(x) \in \mathbb{R}^{2048}$ (global average pooled features)

Text Encoder f_u:

- BioClinicalBERT (BERT pretrained on PubMed + MIMIC notes)
- Input: t (radiology report text, max 512 tokens)
- Output: $u = f_u(t) \in \mathbb{R}^{768}$ ([CLS] token embedding)

Projection Heads:

- Linear layers project to shared 512-dim embedding space
- $z_v = W_v v$, $z_u = W_u u$, where $z_v, z_u \in \mathbb{R}^{512}$
- L2 normalization: $\tilde{z}_v = z_v / \|z_v\|_2$

4.2.2 Training: Contrastive Learning

Given batch of N image-report pairs $\{(x_i, t_i)\}_{i=1}^N$:

Contrastive Loss (Symmetric):

$$\mathcal{L}_{\text{CL}} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{\exp(\tilde{z}_{v,i}^T \tilde{z}_{u,i}/\tau)}{\sum_{j=1}^{N} \exp(\tilde{z}_{v,i}^T \tilde{z}_{u,j}/\tau)} + \log \frac{\exp(\tilde{z}_{u,i}^T \tilde{z}_{v,i}/\tau)}{\sum_{j=1}^{N} \exp(\tilde{z}_{u,j}^T \tilde{z}_{v,i}/\tau)} \right] \tag{1}$$

where $\tau = 0.07$ is temperature parameter.

Training Details:

- Optimizer: Adam, lr $= 5 \times 10^{-5}$
- Batch size: 128 (or smaller with gradient accumulation)
- Epochs: 15-20
- Data augmentation: Random horizontal flip, rotation $\pm 10°$
- Hardware: $1\times$ NVIDIA A100 (24 hours) or V100 (48 hours)

4.2.3 Zero-Shot Classification

For disease d, construct prompt templates:

Positive prompts $\mathcal{T}_+$:

- ”Findings consistent with [disease]”
- ”Evidence of [disease]”
- ”Patient has [disease]”

Negative prompts $\mathcal{T}_-$:

- ”No evidence of [disease]”
- ”Findings not consistent with [disease]”
- ”Normal chest X-ray without [disease]”

Classification score:

$$s_d(x) = \frac{1}{|\mathcal{T}_+|} \sum_{t \in \mathcal{T}_+} \cos(\tilde{z}_v(x), \tilde{z}_u(t)) - \frac{1}{|\mathcal{T}_-|} \sum_{t \in \mathcal{T}_-} \cos(\tilde{z}_v(x), \tilde{z}_u(t)) \tag{2}$$

Decision: $\hat{y}_d = \mathbb{1}[s_d(x) > \theta]$, where θ is threshold (typically 0, or calibrated on validation set).

4.2.4 Expected Baseline Performance

Based on CheXzero paper and our preliminary experiments:

Disease	**IU X-Ray (AUROC)**	**NIH-14 (AUROC)**
Atelectasis	0.78-0.81	0.73-0.76
Cardiomegaly	0.83-0.86	0.80-0.83
Edema	0.88-0.91	0.85-0.88
Pleural Effusion	0.90-0.93	0.87-0.90
Mean (5 diseases)	**0.84-0.87**	**0.81-0.84**

Performance drop on NIH-14 expected due to domain shift (single institution vs. NIH Clinical Center).

4.3 Extension 1: Uncertainty-Aware Classification

4.3.1 Motivation

Clinical deployment requires knowing when model is uncertain. Current CheXzero produces deterministic scores without confidence estimates. Consequences:

- Cannot defer ambiguous cases to human experts
- Overconfident predictions on out-of-distribution data
- Difficult to set operating points balancing sensitivity and specificity

Research Question: Can we augment CheXzero with calibrated uncertainty estimates while maintaining zero-shot performance?

4.3.2 Technical Approach

We propose two complementary methods:

Method 1: Monte Carlo Dropout (MC Dropout)

Idea: Treat dropout as Bayesian approximation. At inference, enable dropout and perform T forward passes:

$$\hat{p}(y_d = 1|x) = \frac{1}{T}\sum_{t=1}^{T}\sigma(s_d^{(t)}(x)) \tag{3}$$

where $s_d^{(t)}(x)$ is classification score with dropout mask t, σ is sigmoid.

Predictive Entropy (Uncertainty):

$$H(y_d|x) = -\hat{p}\log\hat{p} - (1-\hat{p})\log(1-\hat{p}) \tag{4}$$

High entropy $\rightarrow$ model uncertain $\rightarrow$ defer to radiologist.

Implementation:

- Add dropout (p=0.2) after penultimate layer of ResNet-50 and BERT
- At inference: $T = 20$ stochastic forward passes (overhead: 20× single pass)
- Aggregate predictions: mean ± std

Method 2: Conformal Prediction

Idea: Distribution-free confidence sets with coverage guarantees. Given calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}$, construct prediction sets $C(x)$ such that:

$$\mathbb{P}(y \in C(x)) \geq 1 - \alpha \tag{5}$$

for user-specified error rate α (e.g., 0.05 for 95% coverage).

Procedure:

1. Compute non-conformity scores on calibration set: $R_i = |s_d(x_i) - y_i|$
2. Compute quantile: $q = \text{Quantile}(R_1, ..., R_m; 1 - \alpha)$
3. For test image x: predict $\hat{y} \in [s_d(x) - q, s_d(x) + q]$

Advantages:

- Finite-sample coverage guarantee (not asymptotic)
- No distributional assumptions
- Fast calibration (single pass over calibration set)

Disadvantage: Requires held-out calibration set (use validation split).

4.3.3 Evaluation

Metrics:

1. **Calibration:** Expected Calibration Error (ECE)

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \tag{6}$$

where B_m are bins of predicted confidence, $\text{acc}(B_m)$ is accuracy in bin, $\text{conf}(B_m)$ is average confidence.

2. **Coverage:** For conformal prediction, verify empirical coverage $\geq 1 - \alpha$.

3. **Selective Prediction:** Reject fraction f of most uncertain predictions. Plot accuracy vs. f. Goal: steep curve (rejecting few uncertain cases improves accuracy significantly).

4. **Rare Disease Performance:** Test on rare diseases (Pneumothorax: 1.1% in IU X-Ray). Do uncertainty estimates correlate with classification errors?

4.3.4 Expected Contributions

- First application of conformal prediction to medical VLMs
- Demonstrate improved calibration (ECE reduction 20-30%)
- Show 10-15% accuracy improvement by deferring 15% most uncertain cases
- Provide confidence intervals for zero-shot predictions

Publication venue: Radiology AI (clinical utility), MIDL (methodology)

4.4 Extension 2: Factuality-Constrained Report Generation

4.4.1 Motivation

Report generation models frequently hallucinate findings:

- Generate "fracture" when none exists
- Omit critical findings like pneumothorax
- Produce anatomically impossible descriptions

BLEU/ROUGE metrics fail to detect these errors. RadGraph F1 captures factuality but is typically used only for evaluation, not during training.

Research Question: Can we integrate RadGraph-based factuality loss into training to reduce hallucination?

4.4.2 Technical Approach

Base Model: R2Gen

R2Gen architecture:

- **Visual Encoder:** ResNet-101 (grid features) + Vision Transformer (patch features)
- **Memory Module:** M learnable slots $\{m_1, ..., m_M\}$ storing clinical patterns
- **Decoder:** Transformer with cross-attention to visual features and memory

Standard training loss:

$$\mathcal{L}_{\text{CE}} = -\sum_{t=1}^{T} \log p(y_t | y_{<t}, x) \tag{7}$$

Proposed Extension: RadGraph-Based Factuality Loss
Step 1: Extract Entities During Training
For each generated report $\hat{y}$ and reference report y^*:

1. Apply RadGraph IE: $G_{\hat{y}} = \text{RadGraph}(\hat{y})$, $G_{y^*} = \text{RadGraph}(y^*)$
2. G = set of (entity, label) and (entity, relation, entity) tuples

Step 2: Define Factuality Metrics
Precision (Anti-Hallucination):

$$P_{\text{RadGraph}} = \frac{|G_{\hat{y}} \cap G_{y^*}|}{|G_{\hat{y}}|} \tag{8}$$

Recall (Completeness):

$$R_{\text{RadGraph}} = \frac{|G_{\hat{y}} \cap G_{y^*}|}{|G_{y^*}|} \tag{9}$$

F1:

$$F1_{\text{RadGraph}} = \frac{2PR}{P + R} \tag{10}$$

Step 3: Training Objective
Combined Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda(1 - F1_{\text{RadGraph}}) \tag{11}$$

where λ controls factuality weight (tune on validation: $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$).
Challenge: RadGraph extraction is non-differentiable (discrete graph operations).
Solution: Use as auxiliary loss computed per-batch:

1. Generate reports with beam search or sampling
2. Extract RadGraph entities and relations
3. Compute $F1_{\text{RadGraph}}$ on batch
4. Use as reward in policy gradient or as soft constraint

Alternative (Simpler): Train CheXbert classifier as auxiliary task. Add multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{CheXbert}} \tag{12}$$

where $\mathcal{L}_{\text{CheXbert}}$ is binary cross-entropy on 14 CheXpert labels extracted from generated text.

4.4.3 Implementation Plan

Phase 1 (Baseline):

1. Implement R2Gen on IU X-Ray
2. Reproduce BLEU-4 = 0.20 baseline
3. Add RadGraph F1 evaluation (expected: 0.40-0.42)

Phase 2 (Extension):

1. Integrate RadGraph extraction in training loop
2. Experiment with factuality loss variants:
    - Auxiliary CheXbert loss (simpler, differentiable)
    - Policy gradient with RadGraph F1 reward (stronger, harder to train)
    - Retrieval augmentation (retrieve similar reports, condition generation)
3. Hyperparameter tuning: λ, beam size, memory slots

Phase 3 (Evaluation):

1. Quantitative: BLEU, RadGraph F1, CheXbert F1
2. Qualitative: Manual analysis of 100 generated reports
3. Error taxonomy: False positive entities, false negatives, severity errors

4.4.4 Expected Contributions

- Demonstrate RadGraph F1 improvement: 0.42 $\rightarrow$ 0.48 (+6% absolute, +14% relative)
- Reduce false entity rate by 20-30%
- Maintain or slightly improve BLEU (factuality-fluency tradeoff)
- Provide error analysis: which entity types most prone to hallucination?

Publication venue: MICCAI, EMNLP, TMI

4.5 Extension 3: Cross-Dataset Robustness Analysis

4.5.1 Motivation

Most papers train and evaluate on single dataset. External validation rare. Unknown:

- How much does performance degrade on other institutions?
- Which diseases generalize well vs. poorly?
- Are errors systematic (e.g., false positives on rare equipment artifacts)?

Research Question: What is the cross-dataset robustness of medical VLMs, and can simple techniques improve generalization?

4.5.2 Experimental Protocol

Setup:

1. Train CheXzero on IU X-Ray (3,955 studies)
2. Test zero-shot on:
    - IU X-Ray test set (in-distribution)
    - NIH ChestX-ray14 (out-of-distribution)
3. Measure AUROC, AUPRC, F1 per disease
4. Analyze failure modes: false positives, false negatives

Hypotheses:

- **H1:** Performance drops 5-10% AUROC on NIH-14 due to domain shift
- **H2:** Rare diseases (Pneumothorax, Pneumonia) degrade more than common diseases (Cardiomegaly, Effusion)
- **H3:** False positive rate increases on NIH-14 due to dataset-specific artifacts

4.5.3 Domain Adaptation Techniques

Baseline: No adaptation (zero-shot transfer)

Technique 1: Test-Time Augmentation (TTA)

- Apply multiple augmentations (flip, rotate, crop) at test time
- Average predictions: $\hat{y} = \frac{1}{K}\sum_{k=1}^{K} f(x_k)$
- Expected improvement: +1-2% AUROC (free, no retraining)

Technique 2: Few-Shot Fine-Tuning

- Use 1%, 5%, 10% of NIH-14 labeled data
- Fine-tune classification head (freeze encoder or full model)
- Plot: labeled data fraction vs. AUROC recovery

Technique 3: Domain-Invariant Features

- Add domain classifier on intermediate features
- Adversarial training: maximize task performance, minimize domain discrimination
- Loss: $\mathcal{L} = \mathcal{L}_{\text{task}} - \lambda\mathcal{L}_{\text{domain}}$

Technique 4: Ensemble

- Train multiple CheXzero models with different:
  - Random seeds
  - Data augmentation strategies
  - Prompt templates
- Aggregate predictions: majority vote or averaging
- Expected: +2-3% AUROC, better calibration

4.5.4 Evaluation

Quantitative Metrics:

- AUROC, AUPRC per disease
- Mean AUROC across diseases (macro-average)
- Performance drop: $\Delta = \text{AUROC}_{\text{IU}} - \text{AUROC}_{\text{NIH}}$
- Calibration: ECE on both datasets

Qualitative Analysis:

- Error analysis: Sample 100 false positives and false negatives from NIH-14
- Identify patterns:
  - Equipment artifacts (e.g., pacemaker wires misclassified as consolidation)
  - Projection differences (AP vs. PA)
  - Labeling errors in NIH-14 (recall: NLP-extracted labels)
- Visualize: Attention maps on misclassified examples

4.5.5 Expected Contributions

- Comprehensive cross-dataset benchmark:
  - IU X-Ray: 0.85 AUROC
  - NIH-14: 0.78 AUROC (baseline, $\Delta = 7\%$)
  - With few-shot (5%): 0.82 AUROC ($\Delta = 3\%$)
- Identify diseases with poor generalization (rare diseases, subtle findings)
- Demonstrate simple techniques (TTA, few-shot) recover 50-70% of performance drop
- Provide failure taxonomy for future work

Publication venue: MICCAI, MIDL (emphasize robustness and clinical deployment)

4.6 Implementation Timeline

Table 2 breaks down implementation phases:

Table 2: Detailed Implementation Timeline

Phase	Duration	Tasks
Setup	Weeks 1-2	Environment, datasets, reproduce baselines
Extension 1	Weeks 3-6	Uncertainty quantification
Extension 2	Weeks 7-10	Factuality-constrained generation
Extension 3	Weeks 11-14	Cross-dataset evaluation
Analysis	Weeks 15-18	Ablations, comparisons, error analysis
Writing	Weeks 19-24	Paper draft, revision, submission

Risk Mitigation:

- If Extension 2 (factuality loss) proves too complex, focus on Extensions 1 and 3
- Maintain weekly progress logs and pivot if blocked
- Target workshop paper initially (4 pages), extend to conference (8 pages) if results strong